

Takeaways from Former Competition Participant

1. Portfolio Construction & Risk Management

- **Keep turnover low** — around **10% per month** was suggested as a reasonable target.
- Aim for **low correlation with the market** and other unintended systematic exposures.
- Be extremely careful about **look-ahead bias**, including **human look-ahead bias** when manually designing or selecting signals.
- Consider imposing **sector exposure constraints** so that performance is not driven by one concentrated sector bet.
- Account for realistic implementation issues:
 - **Transaction costs**
 - **Slippage**
 - **Short-selling / borrow costs**
 - **Collateral requirements for short positions**
- Think like an actual **portfolio manager**: ask whether the strategy would be acceptable and implementable in a real investment process, not just whether the backtest looks good.

This fits the official competition emphasis quite well. The rules explicitly ask you to report turnover, short-book characteristics, concentration, gross/net exposure and realised beta, and they warn that high turnover can erase alpha through trading costs.

2. Model Design

- The model should be **easy to understand and explain**.
- It does **not have to be simple** — the methodology can be sophisticated — but you should be able to **clearly explain**:
 - what the model is learning,
 - why the inputs should predict returns,
 - and why the resulting signal makes financial sense.
- Avoid relying on a completely opaque **black-box model** whose decisions you cannot defend.
- Creativity can come from either:
 - the **model / algorithm**, or
 - the **investment idea / signal construction**,
 - and ideally from the combination of both.
- **Technical innovation and financial intuition should complement each other.**

This is almost exactly how the official evaluation criteria frame it: sophistication itself is not what is rewarded; they care about “**why this technology, for this signal, on this data.**”

3. Portfolio Optimization Should Be Quantitative

I would rewrite your note:

“Traders have to be pure math — the portfolio part has to be pure math. Don’t use another AI to decide the weights.”

as:

Portfolio construction and position sizing should be systematic and mathematically defined.

Use a transparent optimization or weighting framework rather than allowing an LLM/AI agent to arbitrarily decide portfolio weights.

For example, the model/agent could produce a **forecast or stock score**, but the next step could be something like:

Expected-return scores → optimizer → portfolio weights

subject to constraints such as:

- gross exposure $\leq 200\%$
- net exposure within the allowed band
- beta close to zero
- sector exposure limits
- position-size limits
- turnover budget

Interestingly, the official document *does* mention a “portfolio-construction agent,” but it still expects portfolio construction to operate under explicit constraints and be explainable.

So I wouldn’t interpret the speaker’s comment as “**AI is forbidden in portfolio construction.**” I’d interpret it more as:

Do not let an LLM subjectively choose weights. Make the final allocation rule reproducible and mathematically defensible.

4. Performance Metrics That Matter

Important metrics mentioned:

- **Sharpe Ratio**
- **Information Ratio (IR)** — particularly important
- Alpha
- Market beta / market correlation
- Drawdown
- Turnover

For this year's competition, **Information Ratio is explicitly the headline metric**:

$$IR = \sqrt{12} \times \frac{\text{mean}(\text{active return})}{\text{std}(\text{active return})}$$

where active return means your portfolio return relative to the **3-month T-bill + 4% annual benchmark**.

The official deck requirements also specifically ask for Sharpe, alpha, beta, correlation with the S&P 500, turnover, drawdown and exposure statistics.

5. Look-Ahead Bias — Major Warning

I would give this its own highlighted note:

Look-ahead bias is one of the biggest risks in the competition.

There are really two versions:

Data look-ahead bias:

Using information that was not available when the portfolio decision was made.

Example:

Using a February return or February filing to construct the February 1 portfolio.

Human / model look-ahead bias:

Even if your dataset is clean, *you* may design the strategy after already knowing what worked during the OOS period.

And with LLMs there is another important issue:

A modern LLM may already know historical events from 2021–2026 through its training data.

The official competition document explicitly warns about this and says that asking an LLM for its opinion about a historical company/month may amount to **recall rather than forecasting**.

So a strong pipeline should look more like:

Information available at t

→ **model / signal**

→ **forecast $t+1:t+1$**

- **portfolio optimization**
- **realized t+1t+1 return**

with every stage respecting the same information cutoff.

One thing I would change in your notes: **Sector Neutrality**

Your last note says:

“Does it have to be sector neutral? (<35% ideally)”

Based on the official rules: **No, sector neutrality is not mandatory.**

The document describes **sector or factor neutrality as an “optional refinement.”** The hard constraints are:

- **100–500 total positions**
- **Gross exposure $\leq$ 200%**
- **Net exposure between –50% and +50%**
- Market-neutral mandate, with realised beta expected to remain close to zero.

So I would write your note as:

Sector neutrality is not required, but sector concentration should be controlled. A sector cap — for example around 35% — could be introduced as a portfolio risk constraint if justified.

And I would **not treat 35% as an official number** unless the former participant specifically said it was his recommendation. I don't see a 35% sector cap in the 2026 competition rules.